

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 07: WORK, ENERGY & SIMPLE MACHINES

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-07

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): Scientifically, work is said to be done only when a force produces displacement in the direction of the force.

Reason (R): Work is mathematically defined as the scalar dot product of force and displacement vectors ($W = \mathbf{F} \cdot \mathbf{s} = F s \cos\theta$).

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|---|---|
| <p>(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).</p> <p>(C) Assertion (A) is true but Reason (R) is false.</p> | <p>(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).</p> <p>(D) Assertion (A) is false but Reason (R) is true.</p> |
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Q.2

Assertion (A): A coolie carrying a heavy 50 kg trunk on his head while walking horizontally across a railway platform does zero work against gravity.

Reason (R): The gravitational force acts vertically downward while the displacement is purely horizontal (angle $\theta = 90^\circ$; $\cos 90^\circ = 0$).

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.3

Assertion (A): The work done by the gravitational force on an artificial satellite revolving in a circular orbit around the Earth is zero.

Reason (R): The gravitational centripetal force acts towards the center of the Earth, which is perpendicular to the satellite's instantaneous tangential displacement at every point.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.4

Assertion (A): Work done can be negative.

Reason (R): When a force acts in a direction opposite to the direction of motion (such as kinetic friction or gravity on an upward-moving stone), the angle $\theta = 180^\circ$ and $\cos 180^\circ = -1$.

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|--|--|
| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.5

Assertion (A): The kinetic energy of an object quadruples if its speed is doubled.

Reason (R): Kinetic energy is directly proportional to the square of the speed ($E_k = \frac{1}{2}mv^2$).

- | | |
|--|--|
| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.6

Assertion (A): Energy is a scalar quantity, and its SI unit is the Joule (J).

Reason (R): One Joule is defined as the amount of work done when a force of 1 Newton displaces an object through a distance of 1 meter along the line of action of the force.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.7

Assertion (A): The total mechanical energy of an oscillating simple pendulum remains conserved in the absence of air resistance.

Reason (R): As the pendulum oscillates, potential energy at extreme positions continuously converts into kinetic energy at the mean position and vice versa, keeping $E_k + E_p = \text{constant}$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.8

Assertion (A): Power is defined as the rate of doing work or the rate of transfer of energy.

Reason (R): The SI unit of power is the Watt (W), where 1 Watt equals 1 Joule per second ($1 \text{ W} = 1 \text{ J/s}$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.9

Assertion (A): Commercial electricity consumption is billed in kilowatt-hours (kWh) rather than Joules.

Reason (R): The Joule is an extremely small unit of energy; $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$ (3.6 MegaJoules).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.10

Assertion (A): A compressed spring in a toy gun possesses gravitational potential energy.

Reason (R): Energy possessed by a body due to change in its physical shape or configuration is elastic potential energy.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): When a ball is dropped from a height h , its velocity just before striking the ground is $v = \sqrt{2gh}$, independent of its mass.

Reason (R): According to conservation of mechanical energy, loss in gravitational potential energy equals gain in kinetic energy ($mgh = \frac{1}{2}mv^2$), and mass cancels out.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): A person pushing against a massive rigid concrete wall until exhausted performs zero scientific work on the wall.

Reason (R): Although the person exerts considerable muscular force and burns biochemical calories, the physical displacement of the wall is zero ($s = 0$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): A simple machine like a crowbar or lever can multiply energy.

Reason (R): A machine can multiply force or speed, but according to the law of conservation of energy, work output can never exceed work input.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): In a class 1 lever (such as a crowbar or pair of scissors), the mechanical advantage can be greater than, equal to, or less than 1.

Reason (R): In a class 1 lever, the fulcrum lies between the effort and the load, allowing the effort arm to be longer than, equal to, or shorter than the load arm.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): An electric crane of power 2 kW can lift a load twice as fast as a crane of power 1 kW.

Reason (R): Power is directly proportional to the speed at which a constant force is applied ($P = F \times v$).

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): The efficiency of any real practical machine is always strictly less than 100% (or less than 1).

Reason (R): Some fraction of the input energy is unavoidably dissipated as heat and sound to overcome friction between moving mechanical parts.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): A heavy truck moving at 30 km/h requires four times the braking distance to stop compared to when it moves at 15 km/h under identical braking force.

Reason (R): According to the work-energy theorem, the work done by braking force ($F \times d$) must equal the initial kinetic energy ($\frac{1}{2}mv^2$), so stopping distance $d \propto v^2$.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): When a stone is thrown vertically upward, the work done by gravity during the upward ascent is negative.

Reason (R): The displacement vector is upward while the gravitational force vector acts downward, making the angle between them 180° .

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): A single movable pulley has an ideal mechanical advantage of 2.

Reason (R): In a single movable pulley, the load is supported by two parallel segments of the rope, halving the effort required to lift the load.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): When a fast-moving bullet embeds itself into a stationary wooden block, total mechanical energy is conserved.

Reason (R): In an inelastic collision, total linear momentum is conserved, but mechanical energy is partially converted into internal thermal energy and sound.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): A stretched rubber bow possesses elastic potential energy that can propel an arrow forward.

Reason (R): Work done in deforming the elastic bowstring against restoring intermolecular forces is stored as potential energy.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): If a man walks up a flight of stairs in 10 seconds, he performs more mechanical work than if he walks up the same stairs in 30 seconds.

Reason (R): The mechanical work done against gravity depends solely on the vertical height and body mass ($W = mgh$) and is independent of time.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): In a hydroelectric power station, potential energy of water stored at high elevation is converted into electrical energy.

Reason (R): Stored gravitational potential energy converts to kinetic energy as water falls, which spins turbines connected to electric generators.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): The work done by kinetic friction on a sliding block is always independent of the path taken between two points.

Reason (R): Friction is a non-conservative force whose work depends directly on the total path length traversed.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): A wheelbarrow is an example of a class 2 lever.

Reason (R): In a wheelbarrow, the load is situated between the fulcrum (wheel axle) and the effort (handles), resulting in a mechanical advantage always greater than 1.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(A)	Assertion is true. Reason is true and provides the rigorous vector and trigonometric definition of scientific work.
Q.2	(A)	Assertion is true. Reason is true and explains why orthogonal force and displacement yield zero gravitational work.
Q.3	(A)	Assertion is true. Reason is true and proves why centripetal force does zero work in circular orbits.
Q.4	(A)	Assertion is true. Reason is true and mathematically justifies negative work ($W = -F \cdot s$).
Q.5	(A)	Assertion is true. Reason is true and explains the quadratic relationship between velocity and kinetic energy.
Q.6	(A)	Assertion is true. Reason is true and gives the standard SI definition of 1 Joule.
Q.7	(A)	Assertion is true. Reason is true and details the periodic interconversion of kinetic and potential energy.
Q.8	(A)	Assertion is true. Reason is true and provides both definition and SI unit equivalency.
Q.9	(A)	Assertion is true. Reason is true and explains the practical engineering necessity for adopting the larger unit kWh.
Q.10	(D)	Assertion is false: A compressed spring possesses elastic potential energy, not gravitational potential energy. Reason is true.
Q.11	(A)	Assertion is true. Reason is true and shows algebraic proof of velocity upon impact.
Q.12	(A)	Assertion is true. Reason is true and highlights the strict physical condition $W = F \cdot 0 = 0$.
Q.13	(D)	Assertion is false: No machine can multiply energy or violate energy conservation. Reason is true (machines multiply force, not energy).
Q.14	(A)	Assertion is true. Reason is true and provides the structural mechanics of Class 1 levers.
Q.15	(A)	Assertion is true. Reason is true and provides the relation $P = Fv$, showing lifting speed doubles with doubled power.
Q.16	(A)	Assertion is true. Reason is true and explains the thermodynamic irreversibility caused by friction in machines.
Q.17	(A)	Assertion is true ($2^2 = 4$). Reason is true and gives the work-energy theorem derivation for braking distance.
Q.18	(A)	Assertion is true. Reason is true and explains why upward gravitational work is negative.
Q.19	(A)	Assertion is true. Reason is true and describes how tension in two supporting segments yields $\text{MA} = 2$.
Q.20	(D)	Assertion is false: Mechanical energy is not conserved (kinetic energy decreases due to thermal dissipation). Reason is true.
Q.21	(A)	Assertion is true. Reason is true and explains how deformation work is transformed into elastic potential energy.
Q.22	(D)	Assertion is false: Mechanical work done is identical (mgh); only his power output is greater in 10 seconds. Reason is true.
Q.23	(A)	Assertion is true. Reason is true and traces the multi-step energy transformation in hydroelectric dams.
Q.24	(D)	Assertion is false: Friction work depends on the path (longer path = more work dissipated). Reason is true (friction is non-conservative).
Q.25	(A)	Assertion is true. Reason is true and correctly classifies the wheelbarrow as a force-multiplying Class 2 lever.